

Week 4: Injective functions

or "one-to-one"

Defn: We say that a function $f: A \rightarrow B$ is injective if

$$\forall a, b \in A, a \neq b \Rightarrow f(a) \neq f(b)$$

Slogan: "Distinct inputs give distinct outputs"

Contrapositive: $\forall a, b \in A, f(a) = f(b) \Rightarrow a = b$

(often easier to do proofs with)

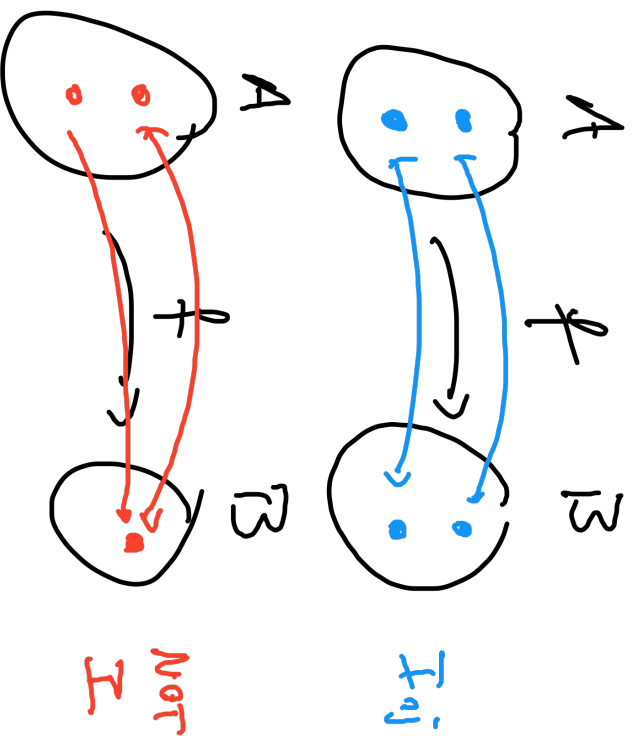
Regret: $\exists a, b \in A$ s.t. $a \neq b$ AND $f(a) = f(b)$

#1 Mistake:

$$\forall a, b \in A, a = b \Rightarrow f(a) = f(b)$$

wrong

just ... the defn of a fcn



"Test functions"



f_{a1}



f_{a1}

$a \neq b \text{ and } f(a) = f(b)$

Examples:

$$g: \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x^2$$

NOT INJ

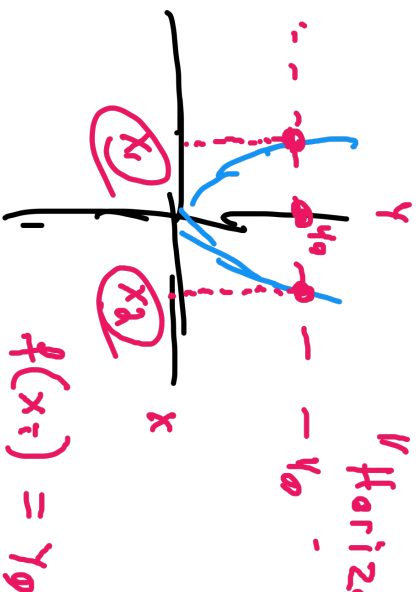
$1 \neq -1$ and

$$g(1) = g(-1)$$

$$''$$

$$''$$

$$''$$



$\mathbb{R} \rightarrow \mathbb{R}$

"Horizon. line test"?

A Hor. line L , the intersects
of L with the graph is
at most one point

(2.7)

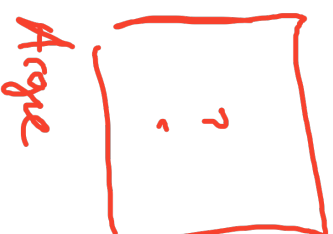
f is inj.

Note: to prove a function $f: A \rightarrow B$ is injective, need an argument,

$$\text{"} \forall a, b \in A, a \neq b \Rightarrow f(a) \neq f(b) \text{"}$$

$$\text{"} \forall a, b \in A, f(a) = f(b) \Rightarrow a = b \text{"}$$

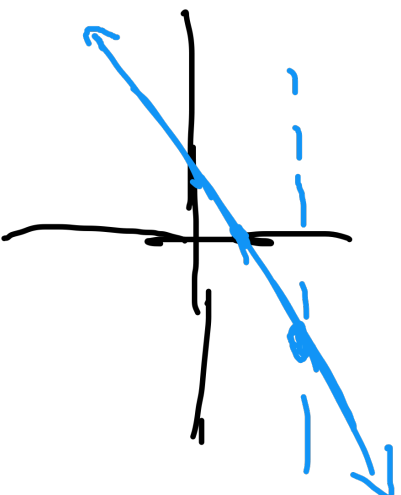
"Let $a, b \in A$. Assume $f(a) = f(b)$.



... We conclude "
that $a = b$.

Example: $\mathbb{R} \xrightarrow{f} \mathbb{R}$

$x \mapsto \frac{x+1}{2}$



Claim: f is inj,

Proof: let $a, b \in \mathbb{R}$. Suppose $f(a) = f(b)$. IE,

$\frac{a+1}{2} = \frac{b+1}{2}$. (WTS: $a=b$) Multiplying by 2 gives $a+1 = b+1$.

Subtracting 1 gives $a=b$. $\square$

$$\underline{\text{Ei}} \quad \mathbb{R}_{>0} \xrightarrow{f} \mathbb{R} \quad \textcircled{\text{I}}$$

$$x \longmapsto \frac{x-1}{x+1}$$

Proof: let $a, b \in \mathbb{R}_{>0}$. Suppose $f(a) = f(b)$, i.e., $\frac{a-1}{a+1} = \frac{b-1}{b+1}$.

Clearing denominators gives $(a-1)(b+1) = (b-1)(a+1)$, i.e.,

$$ab - b + a - 1 = ab - a + b - 1, \text{ Thus } -b + a = -a + b, \text{ so } 2a = 2b, \text{ Thus } a = b. \quad \square$$

$$\frac{x-1}{x+1} = \frac{x+1-a}{x+1} = 1 - \frac{a}{x+1} \Rightarrow 1 - \frac{a}{a+1} = 1 - \frac{a}{b+1}$$

$$\mathbb{R}^3 \xrightarrow{g} \mathbb{R}^2$$

$$(x, y, z) \mapsto (x, y)$$

$$g(1, 2, 3) = (1, 2)$$

$$g(1, 2, 4) = (1, 2)$$

$$\text{but } (1, 2, 3) \neq (1, 2, 4)$$

g is NOT inj

$$\mathbb{R}^2 \xrightarrow{f} \mathbb{R}^3 \quad \boxed{+}$$

$$(x, y) \longmapsto (x+y, x-y, x^2+y^2)$$

$$(0, 0) \longmapsto (0, 0, 0)$$

$$(1, 1) \longmapsto (2, 0, 2)$$

⋮

Proof: SpS $(0, b), (c, d) \in \mathbb{R}^2$. SpS $f(a, b) = f(c, d)$.

$$\text{Thus } (a+b, a-b, a^2+b^2) = (c+d, c-d, c^2+d^2).$$

$$\text{Thus } a+b = c+d, \quad a-b = c-d, \quad a^2+b^2 = c^2+d^2.$$

Adding the 1st + 2nd eqn gives $2a = 2c$, so $a = c$.

~~Subtracting~~ Subtracting gives $2b = 2d$, thus $b = d$.

We conclude that

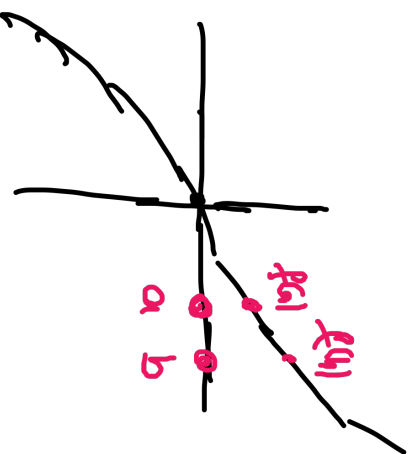
$$(a, b) = (c, d) \quad \square$$

$$\begin{array}{c} \text{wTS} \\ (a, b) = (c, d) \\ \text{wTS} \\ a = c \quad b = d \end{array}$$

Example: $\mathbb{R} \rightarrow \mathbb{R}$

$$x \mapsto x^3 + x$$

$$x^3 + x = x(x^2 + 1)$$



or decreasing $\Rightarrow$ increasing $\Rightarrow$ increasing

$$b > a \Rightarrow f(b) > f(a)$$

$a \neq b$ + f increasing

$$a < b \Rightarrow f(a) < f(b) \Rightarrow f(a) \neq f(b)$$

$$a > b \Rightarrow f(a) > f(b) \Rightarrow f(a) \neq f(b).$$

Try "usual" proof: let $a, b \in \mathbb{R}$. s.t. $f(a) = f(b)$. Then

$$a^3 + a = b^3 + b \dots$$

f cts, diff. *Prove to go...*

Calc I: $f'(x) > 0 \Rightarrow f$ is increasing

Proof: Since $f'(x) = 3x^2 + 1$, $f'(x) \geq 1 \forall x \in \mathbb{R}$. Thus f is increasing and therefore injective. $\square$

Ex's:

$$f(x) = x^5 + 7x^3 + 3x$$

$$f'(x) = \boxed{5x^4 + 7x^2 + 3 \geq 3}$$

$$\Rightarrow f \text{ is inc. } \uparrow$$

Might need

a small argument

to show $f' > 0$.

Example

$$\mathbb{R} \xrightarrow{\cos} \mathbb{R}$$

$$x \mapsto \cos x$$

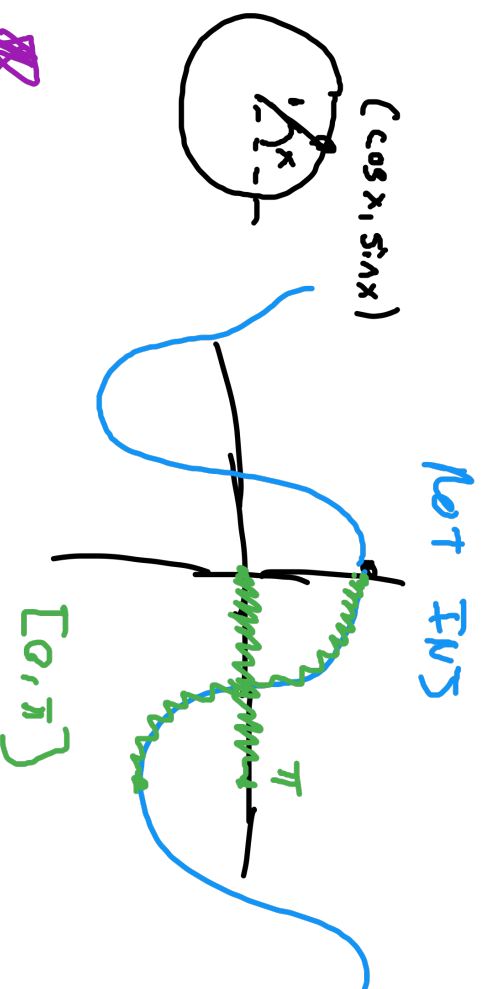
$$\cos 0 = \cos 2\pi$$

$$\text{but } 0 \neq 2\pi.$$

$$[0, \pi] \rightarrow \mathbb{R}$$

$$x \mapsto \cos x$$

is injective.



Different fns b/c different domains

Pft: The der. of $\cos x$ is $-\sin x$. Since $-\sin x \leq 0 \forall x \in [0, \pi]$,

$\cos$ is decreasing, and thus injective. $\square$

$$\begin{array}{ll} \mathbb{C} \mapsto \mathbb{C} & \text{NI } (i)^2 = (-i)^2 \\ x \mapsto x^2 & \text{" } (-1)^2 = 1 \\ & \text{" } (i)^2 = -1 \\ & \text{" } (-i)^2 = -1 \end{array}$$

$$\mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{Z})$$

$$S \mapsto S \cap \mathbb{Z}$$

Not I: Are there sets $S_1, S_2 \subseteq \mathbb{R}$
 s.t. $S_1 \neq S_2$ but $S_1 \cap \mathbb{Z} = S_2 \cap \mathbb{Z}$?

$$\boxed{\{1, \pi\}} \cap \mathbb{Z} = \{1\} = \boxed{\{1, e\}} \cap \mathbb{Z}$$

$$A \xrightarrow{f} B$$

$$x, y \in A$$

Recall: " $f(x) \leq f(y) \implies x \leq y$ " is false.



Counter example
~~" $f(x) \leq f(y) \implies x \leq y$ "~~
 let $A = \{1, 2, 3, \dots\}$

"Find the 'correct' hypothesis"

Not FVS.

L: Sps f is injective and $f(x) \leq f(y)$. Then $x \leq y$.

THE DEF OF
 $\downarrow$
 $f(x)$

P: Sps f is inj and $f(x) \leq f(y)$. Let $a \in X$. Then $f(a) \in f(X)$.

(To use the hypothesis " $f(x) \leq f(y)$ ", need an elt of $f(X)$)

Since $f(a) \in f(X)$, and $f(x) \leq f(y)$, $\boxed{f(a)} \in f(y)$.

Thus $\exists c \in Y$ s.t. $f(a) = f(c)$. Since f is

injective, $a = c$. Since $c \in Y$. $\square$

$c \in f(y) \implies$
 $\exists b \in Y$ s.t.
 $c = f(b)$
 Apply w/ $c = f(a)$